

Neutron Transport and Tritium Breeding Modelling in Nuclear Fusion Reactor Breeder Blankets

MPhys Project Report

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Abstract

The realisation of commercial nuclear fusion via the Deuterium-Tritium (D-T) fuel cycle is contingent upon achieving tritium self-sufficiency. This project utilises a coupled computational workflow, integrating Paramak for parametric CAD generation and OpenMC for stochastic neutron transport, to optimise the neutronic performance of tokamak breeder blankets. A systematic parameter sweep was conducted to evaluate the impact of material composition, isotopic enrichment, and reactor geometry on the Tritium Breeding Ratio (TBR).

Results indicate that the liquid eutectic $\text{Li}_{17}\text{Pb}_{83}$ offers superior performance, achieving a maximum TBR of 1.61, significantly outperforming molten salt and solid ceramic concepts. Contrary to high-enrichment assumptions, this study identifies an enrichment saturation point at approximately 40% ^6Li , beyond which performance diminishes due to the loss of ^7Li neutron multiplication. Geometric optimisation reveals that TBR saturates at a radial blanket depth of 500–600 mm, determined by the neutron mean free path. Under a homogenous approximation of the breeder blanket structure, water coolant was found to enhance breeding via moderation, gaseous coolants (He , N_2) act as voids that degrade the neutron economy. These findings suggest that optimal self-sufficiency is achieved through moderate isotopic enrichment and compact radial builds, directly informing design constraints for next-generation devices.

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1. INTRODUCTION

The global demand for energy is rising inexorably, coinciding with an urgent necessity to decarbonise power generation. In this context, nuclear fusion offers the prospect of a near-limitless, low-carbon energy source. By replicating the stellar nucleosynthesis processes that power the Sun, fusion energy aims to generate electricity without long-lived high-activity radioactive waste or the risk of runaway chain reactions [1].

Among the various reaction pathways, the Deuterium-Tritium (D-T) fuel cycle is the most viable for near-term applications due to its high reaction cross-section at achievable plasma temperatures [2]. However, while deuterium is abundant in earth’s oceans, the practical realisation of this cycle faces a fundamental logistical challenge: the supply of tritium. Tritium is a radioactive isotope ($\tau_{1/2} \approx 12.3$ years) with no significant natural abundance [3]. Current global inventories are derived as by-products of CANDU fission reactors and are insufficient to support a future fleet of commercial fusion power plants [4].

Consequently, a D-T fusion reactor cannot rely on an external fuel supply; it must breed its own fuel. This leads to the concept of the *Breeder Blanket*—a component designed to intercept the 14.1 MeV neutrons produced in the fusion reaction and transmute lithium into tritium. While the blanket serves integral roles in heat extraction and shielding, its defining function—and the primary figure of merit for this study—is the **Tritium Breeding Ratio (TBR)**: the ratio of tritium produced to tritium consumed.

Because the D-T reaction produces only one neutron per consumption event, this “one-to-one” neutron economy would require breeding efficiency (no losses of neutrons / tritium). Real world breeding inefficiencies (see Section 3.2) necessitate achieving a TBR greater than 1.0. As the source of fuel for fusion, tritium breeding is not merely a design objective but a threshold requirement for the viability and scalability of fusion energy [5].

This strict margin necessitates high-fidelity modelling. Experimental validation of full-scale reactor neutronics is currently unfeasible, placing the burden of design on computational simulation. This project applies advanced Monte Carlo transport methods to perform a systematic optimisation of the breeder blanket. By utilising a parametric computational workflow, this study aims to identify the specific material compositions and geometric configurations required to maximise the TBR, ensuring fuel self-sufficiency while maintaining adequate performance for other aspects of fusion energy generation. The outcomes of this work are intended to inform the design of next-generation concepts relevant to future magnetic confinement devices, such as the EU-DEMO (DEMONstration power plant) [6] and the UK STEP (Spherical Tokamak for Energy Production) programme [7].

2. THEORY

2.1. Fundamentals of Nuclear Fusion

Nuclear fusion is the process by which two light atomic nuclei combine to form a heavier nucleus, releasing energy due to the increase in total nuclear binding energy [1]. While various light nuclei interactions can release energy via this mass defect, the deuterium-tritium (D-T) reaction has emerged as the most viable for near-term applications, largely due to historical precedent and physical accessibility [8].

This reaction is favoured because it possesses a significantly higher fusion cross-section (σ_f) at achievable plasma temperatures (10–20 keV) compared to alternatives such as D-D or D- ^3He (see Figure 1) [2]. The reaction proceeds as:



The release of 17.6 MeV is partitioned inversely to the mass of the products to conserve momentum. Consequently, the electrically neutral neutron carries the majority of the energy ($E_n = 14.1$ MeV), while the charged alpha particle (α) retains 3.5 MeV [2].

In Magnetic Confinement Fusion (MCF) devices, such as the tokamak [9], the charged α -particle is confined by the magnetic field, depositing its energy to sustain the plasma temperature (self-heating). The 14.1 MeV neutron, being electrically neutral, escapes the magnetic confinement. The reactor breeder blanket is the subsystem designed to intercept this neutron flux to extract thermal energy, shield the external components, and breed fuel [10].

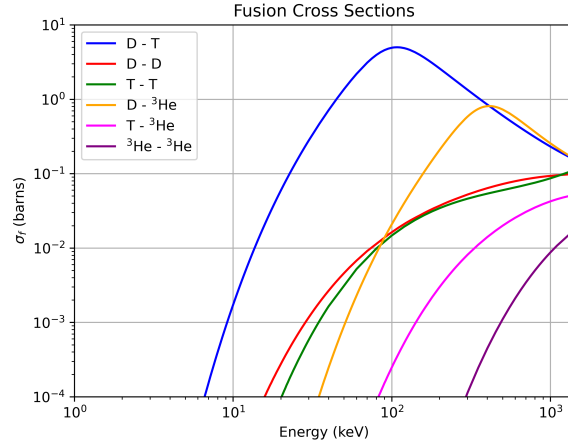


Figure 1: A comparison of fusion reaction reactivities. The total cross-section is plotted as a function of ion temperature (keV) for the D-T, D-D, and D- ^3He fuel cycles. The D-T reaction peaks at a lower temperature and higher cross-section than alternatives [2].

2.2. Tritium Breeding and Neutron Multiplication

The necessity of tritium production was established in Section 1. To achieve this, the blanket must breed tritium via neutron capture interactions with lithium. Natural lithium consists of two isotopes: ^6Li (7.59% abundance) and ^7Li (92.41% abundance) [3].

2.2.1. Breeding Reactions

Both isotopes undergo tritium-producing reactions, but their neutronic characteristics differ fundamentally:



The ^6Li reaction (Eq. 2) is exothermic and possesses a $1/v$ cross-section dependence, making it dominant at thermal neutron energies ($E < 0.5$ eV) (Figure 2) [11]. Conversely, the ^7Li reaction (Eq. 3) is endothermic with a threshold energy of approximately 2.5 MeV [10].

2.2.2. Neutron Multiplication and Economy

The reactor's fuel self-sufficiency is quantified by the **Tritium Breeding Ratio (TBR)**, defined as the ratio of tritium bred to tritium consumed.

However, the D-T reaction produces exactly one neutron, and the primary ^6Li breeding reaction consumes one neutron. This 1-to-1 economy leaves no margin for breeding inefficiency (see

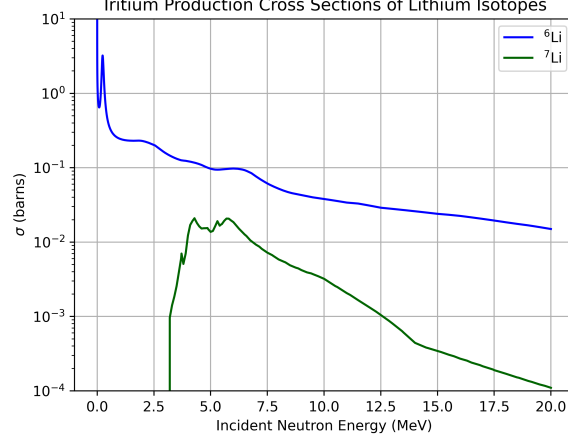


Figure 2: Neutron cross-sections for the primary tritium breeding reactions, ${}^6\text{Li}(n,\alpha)t$ and ${}^7\text{Li}(n, n'\alpha)t$, as a function of incident neutron energy. Note the dominance of ${}^6\text{Li}$ at thermal energies.

Section 3.2). To overcome this deficit, the blanket must utilise neutron multipliers—materials with significant $(n, 2n)$ cross-sections at 14.1 MeV. The primary candidates are Beryllium and Lead, the latter often used in liquid lead-lithium eutectic blankets [12] [3]:



These are threshold reactions (see Figure 3) that increase the macroscopic neutron population, allowing a single incident fusion neutron to potentially breed more than one triton [11].

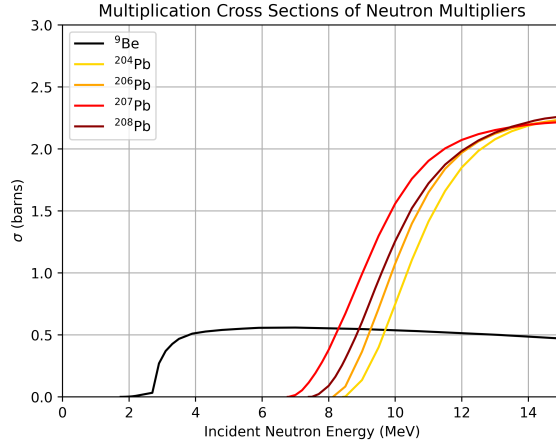


Figure 3: Neutron cross-sections for the $(n, 2n)$ multiplication reactions in Beryllium and Lead. These threshold reactions effectively double the neutron population available for breeding.

2.3. Neutron Transport and Interactions

To determine the TBR, one must model the full phase-space behaviour of neutrons within the complex reactor geometry. This behaviour is governed by the Boltzmann Transport Equation (BTE).

2.3.1. The Boltzmann Transport Equation

The BTE is a linear integro-differential equation describing the conservation of neutrons in phase space $(\vec{r}, \hat{\Omega}, E, t)$. It stipulates that the rate of change of neutron density is equal to the net production minus net loss and net leakage. For a steady-state system with a fixed source, the BTE is given by [13]:

$$\underbrace{\hat{\Omega} \cdot \nabla \psi}_{\text{Streaming}} + \underbrace{\Sigma_t \psi}_{\text{Collision Loss}} = \underbrace{\int_0^\infty \int_{4\pi} \Sigma_s(E' \rightarrow E, \hat{\Omega}' \rightarrow \hat{\Omega}) \psi' d\hat{\Omega}' dE'}_{\text{Backscattering Source}} + \underbrace{S(\vec{r}, \hat{\Omega}, E)}_{\text{Fixed Source}} \quad (6)$$

Where $\psi(\vec{r}, \hat{\Omega}, E)$ is the angular neutron flux, Σ_t is the total macroscopic cross-section, and Σ_s represents the probability of a neutron scattering from energy E' and direction $\hat{\Omega}'$ into E and $\hat{\Omega}$.

Analytic solutions to Eq. 6 are generally impossible for heterogeneous 3D reactor geometries. Consequently, this project employs Monte Carlo (MC) methods. MC does not solve the equation for average flux directly; instead, it simulates the stochastic histories of individual particles, sampling interaction probabilities from nuclear data libraries such as ENDF/B-VIII.0 [3]. The statistical aggregation of these histories converges to the solution of the integral BTE [13].

2.3.2. Interaction Probabilities, Mean Free Path and Moderation

The probability of a specific interaction type x (e.g., capture, fission, scatter) occurring is defined by the macroscopic cross-section Σ_x , which relates to the microscopic cross-section σ_x and atomic density N :

$$\Sigma_x(E) = N\sigma_x(E) \quad (7)$$

The mean free path (λ), representing the average distance a neutron travels through the blanket medium before interacting [1], is:

$$\lambda(E) = \frac{1}{\Sigma_{total}(E)} \quad (8)$$

In blanket design, λ dictates the necessary radial build thickness required to attenuate the 14.1 MeV flux and protect against neutron leakage [14].

Fast fusion neutrons can be moderated, often all the way to thermal energies, using appropriate low mass number materials (such as water). This is detailed further in Appendix A. This maximises the cross section of the breeding reaction 2, and can reduce neutron leakage by attenuating neutron flux to thermal energies.

3. LITERATURE REVIEW

3.1. The Tritium Fuel Imperative

The deuterium-tritium (D-T) fuel cycle has been identified as the most feasible pathway for near-term fusion energy, primarily due to its high reaction cross-section at achievable plasma temperatures [8, 2]. However, this reliance creates a fundamental logistical challenge: the fuel cycle is not self-contained. While deuterium is abundant, tritium is a radioactive isotope with a half-life of 12.3 years and no significant natural abundance.

Current global tritium stockpiles are derived almost exclusively as a by-product of heavy-water fission reactors (CANDU). Analyses of this supply chain indicate that stockpiles are insufficient to support the operation of a future fusion fleet [4, 15]. Consequently, the external supply chain is considered a bottleneck that threatens the viability of the entire D-T fusion

concept. To resolve this, commercial reactors are required to breed their own fuel in-situ via the **Tritium Breeder Blanket**.

3.2. Breeder Blanket Evolution and Functions

The concept of a “breeder” blanket predates magnetic confinement fusion; it was originally conceived to utilise high-energy neutrons to produce fissile materials (^{233}U , ^{239}Pu) for fission warheads and reactors [16, 17, 18]. Modern fusion research has repurposed this architecture to focus on the transmutation of lithium into tritium. Beyond breeding, the blanket must simultaneously perform three conflicting functions: shielding superconducting magnets and external components from neutron damage / radiation, extracting high-grade heat for power conversion, and maintaining a Tritium Breeding Ratio (TBR) > 1.0 [10].

TBR must be > 1.0 give margin for several different sources of inefficiency. Neutrons can be lost to structural materials (e.g. first walls) via parasitic absorption and neutron can also leak out of the reactor vessel. Produced tritium can be lost from processing inefficiencies and tritium’s short half life leads to decay losses. Accounting for these breeding inefficiencies literature typically suggests a target TBR ≈ 1.1 [10, 5, 19]. Achieving this margin is non-trivial; the 1-to-1 neutron economy of the D-T reaction leaves no surplus for , necessitating the use of neutron multipliers to enhance the neutron population [11]. When additionally accounting for decay and processing inefficiencies, a TBR significantly > 1.0 (typically ≈ 1.1) is mandatory for a viable power plant [5, 19, 20].

3.3. Neutron Multiplication Strategies

Since natural lithium cannot sustain a TBR > 1 in a realistic reactor geometry solely via the endothermic $^7\text{Li}(n, n'\alpha)t$ reaction, design studies have converged on two primary multiplier candidates: Beryllium and Lead [12].

Beryllium (^9Be) is neutronicallly superior due to its low threshold energy for the $(n, 2n)$ reaction. It is the multiplier of choice for solid ceramic breeder concepts (e.g., HCPB, WCPB) and molten fluoride salts (FLiBe) [21, 14, 22]. However, its application is controversial due to resource scarcity and high toxicity. Shimwell et al. [23] highlight that the handling of toxic beryllium dust poses severe safety and manufacturing challenges, prompting research into minimizing its volume fraction in blanket designs.

Conversely, Lead (Pb) is favored for its chemical compatibility with liquid breeders, specifically in the Lithium-Lead eutectic ($\text{Li}_{17}\text{Pb}_{83}$). This alloy allows the medium to function simultaneously as breeder, multiplier, and coolant [24]. While lead mitigates the toxicity concerns of beryllium, it introduces neutronic penalties; it acts as a high-Z shield, parasitically absorbing the thermal neutrons required for efficient ^6Li breeding. Furthermore, the activation of lead can generate long-lived radioisotopes, complicating decommissioning [25]. The selection of a multiplier therefore represents a trade-off between the superior neutronic efficiency of beryllium and the integrated engineering advantages of lead [11].

3.4. Divergence in Blanket Architectures

Blanket designs have historically diverged into two competing pathways—Liquid and Solid—driven by the phase of the breeding medium.

3.4.1. Liquid Breeder Concepts

Liquid breeders offer the distinct advantage of continuous tritium extraction and simplified fuel replenishment [14].

3.4.1.1. *Lithium-Lead (Li–Pb):*

The Dual-Cooled Lithium-Lead (DCLL) blanket is a mature concept for the European DEMO [26]. However, the flow of liquid metal through the tokamak’s strong magnetic field induces significant Magnetohydrodynamic (MHD) drag, which can inhibit circulation and heat transfer [12].

3.4.1.2. *Molten Salts:*

To bypass MHD issues, designs such as the ARC reactor utilise molten salts like FLiBe ($2\text{LiF} \cdot \text{BeF}_2$) which possess low electrical conductivity [22, 27]. While solving the MHD problem, these salts introduce new challenges regarding chemical corrosion and complex flow chemistry control [25, 28].

3.4.2. *Solid Breeder Concepts*

Solid breeders utilise lithium ceramics (e.g., Li_2TiO_3 , Li_4SiO_4) in packed pebble beds. This approach eliminates MHD effects entirely and ensures chemical stability [29, 30]. It is the preferred approach for the Chinese CFETR (Water-Cooled Pebble Bed) and Japanese DEMO designs [31, 32]. The primary drawback of solid breeders lies in the complexity of tritium extraction; tritium produced within the ceramic lattice must diffuse out and be purged by a sweep gas, a process sensitive to radiation-induced trapping [33].

3.5. Neutronics Modelling and the Research Gap

Optimising these complex designs requires high-fidelity modelling, as experimental validation of full-scale blanket neutronics is currently impossible [34]. Consequently, the field relies on Monte Carlo (MC) transport codes (e.g., OpenMC, MCNP) to resolve the stochastic behaviour of neutrons in 3D geometries [35, 36].

However, a critical review of the literature reveals a methodological gap in how these tools are applied. Previous studies generally fall into one of two categories:

1. **Broad Material Surveys:** Reviews such as [11] provide excellent “first principles” comparisons of many materials but rely on simplified 1D or homogenised models. These studies sacrifice geometric accuracy for breadth, failing to capture spatial transport effects like streaming or leakage.
2. **Specific Point-Designs:** Studies focused on specific reactors (e.g., ARC [37] or STEP [38]) utilize high-fidelity 3D CAD models but fix the geometry to a single design point. These studies answer “What is the performance of *this* reactor?” rather than “How does geometry influence performance?”

This dichotomy exists because manually generating unique 3D CAD models for a parametric sweep has historically been computationally intractable. Recent developments in parametric CAD generation have begun to bridge this gap. Goel et al. [39] demonstrated the viability of automated geometry generation for neutronic sweeps. However, their study—representing an initial step in this domain—did not fully exploit the coupling between material composition and geometric variance.

This project aims to close this gap by applying an automated, parametric workflow to a realistic 3D tokamak model. Unlike previous studies which isolated material choice from geometric design, this work will conduct a systematic parameter sweep of variables—including varying ^6Li enrichment, blanket thickness, and material ratios. This approach allows for the identification of optimal design points that lie at the intersection of material science and reactor geometry, regions of the design space that effectively remain unmapped in current literature.

4. METHODOLOGY

4.1. Modelling Neutron Transport: From Boltzmann to Monte Carlo

The neutronic performance of the breeder blanket is determined by the transport behaviour of the entire neutron population within the reactor geometry. This behaviour is formally governed by the **Boltzmann Transport Equation (BTE)**, a linear integro-differential equation representing particle balance in phase space [13].

The BTE stipulates that the time rate of change of the neutron density is equal to the net production (via sources and fission/multiplication) minus the net loss (via absorption and leakage) and the net scattering transfer into or out of the phase space volume. While the BTE provides a comprehensive deterministic description, it is computationally intractable to solve analytically for the complex, heterogeneous 3D geometries inherent to fusion reactors. Consequently, this project employs the **Monte Carlo (MC) method**.

The MC method is a stochastic technique that simulates the transport of neutrons by tracking the individual life histories of discrete particles [40]. Rather than solving for global flux directly, the simulation proceeds through a random walk process:

1. **Source Sampling:** A neutron is initialized at the plasma source with sampled parameters (e.g., 14.1 MeV energy and isotropic direction).
2. **Ray Tracing:** The distance to the next interaction is sampled from a probability distribution derived from the total macroscopic cross-section of the local material [3].
3. **Interaction Physics:** At the collision site, the specific reaction type (elastic scatter, absorption, (n, 2n) multiplication, etc.) is selected probabilistically based on the nuclide-specific cross-section data.
4. **History Termination:** The particle’s energy and direction are updated after scattering. The process repeats until the neutron is absorbed or leaks from the system geometry. Multiplication events generate secondary neutrons which are subsequently tracked.

4.1.1. Statistical Uncertainty and Error Analysis

Unlike deterministic solvers, Monte Carlo results represent statistical estimates rather than exact solutions. The physical quantities of interest, such as the Tritium Breeding Ratio (TBR), are tallied by averaging contributions across many independent particle histories. The uncertainty in these results is quantified using the standard deviation of the mean across simulation batches.

The relative error (R) in an MC calculation is governed by the Central Limit Theorem and scales inversely with the square root of the number of simulated particles [13, 35] (N):

$$R \propto \frac{1}{\sqrt{N}} \quad (9)$$

To ensure statistical robustness, the simulations in this work utilised high particle counts (typically $N \sim 10^5 - 10^7$) to drive convergence. This resulted in relative errors significantly below 1%, rendering the associated error bars smaller than the data markers in the presented graphical results.

4.2. Computational Workflow and Toolchain

As established in Section 3, modern neutronics research relies on a robust toolchain linking geometric modeling with particle transport simulation. This project employs a coupled workflow utilising two primary open-source tools: **Paramak** for geometry generation and **OpenMC** for neutron transport.

4.2.1. Geometry Generation: Paramak

To manage the complex toroidal geometry of the tokamak, this project utilises **Paramak**, a Python-based parametric CAD framework [41]. Paramak allows for the automated generation of tokamak fusion reactor models through a high-level API, enabling the rapid modification of geometric features—such as blanket thickness, triangularity, and radial build—without manual CAD reconstruction.

The generated geometry is exported as a DAGMC (Direct Accelerated Geometry Monte Carlo) unstructured mesh. This format is optimised for ray-tracing operations, allowing the Monte Carlo code to transport particles directly on the CAD surface representation without voxelization.

4.2.2. Neutron Transport: OpenMC

All neutronic simulations are performed using **OpenMC**, a state-of-the-art Monte Carlo transport code [35]. OpenMC is widely adopted within the fusion community due to its continuous-energy physics and efficient parallelisation. The code has been extensively validated for complex fusion applications, demonstrating high fidelity in calculating critical parameters such as the Tritium Breeding Ratio (TBR) and volumetric nuclear heating [36].

To ensure computational efficiency while maintaining physical accuracy, the simulations utilise a 90° sectoral revolution of the tokamak with reflective periodic boundary conditions applied to the radial surfaces (Figure 4).

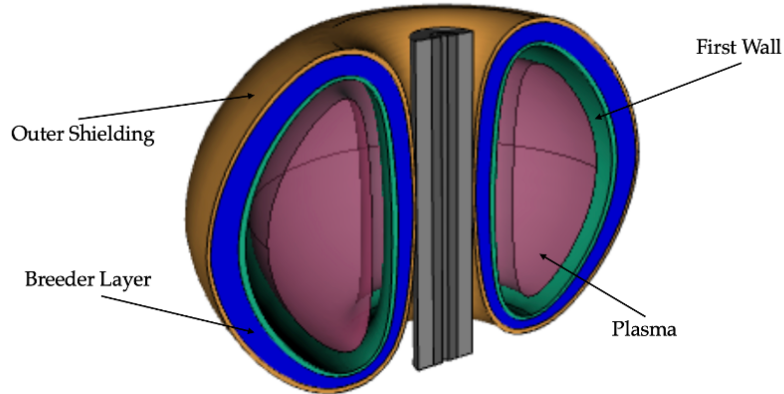


Figure 4: Representative geometry of the simplified Tokamak model generated via Paramak. The simulation domain comprises a 90° revolution with periodic boundary conditions to approximate the full torus while reducing computational cost.

4.3. Definition of the Base Reactor Model

Before undertaking parametric optimisation, a “Base Model” was established to serve as a reference point. The parameters for this baseline were synthesised from literature to ensure the reactor possesses dimensions and material compositions representative of a realistic DEMO-class machine. A summary of these base parameters is provided below.

4.3.1. Neutron Source Specification

The plasma source is modelled as an isotropic, 14.1 MeV neutron ring source located at the magnetic axis. To approximate the Doppler broadening resulting from the thermal motion of plasma ions, the energy spectrum follows a Muir distribution [42]. The ion temperature is set

to 20 keV, consistent with burning plasma conditions described by Wandelt [43]. Note that the plasma is treated solely as a neutron ring source; a physical plasma volume with self and wall interactions are outside the scope of this neutronic study.

4.3.2. *Breeder and Coolant Composition*

The reference liquid breeder material is the eutectic Lithium-Lead alloy ($\text{Li}_{17}\text{Pb}_{83}$). This specific stoichiometry is selected for its low melting point, mitigating freezing concerns in the loop [44]. There is no hard consensus regarding Li_6 enrichment, literature varies from a low as 10% all the way to 90% [11, 45, 27, 37, 46]. To take an average values of the most common values found in breeding studies, the ^6Li enrichment is initialised at 60%.

Water (H_2O) is selected as the base coolant due to its prevalence in designs such as the ITER TBM and DEMO concepts [47]. Coolant Piping Setups differ from significantly between reactor design. To maintain computational tractability, the breeder and coolant are modelled as a homogenised mixture rather than discrete cooling channels. The coolant occupies 33% of the total blanket volume, as an estimation of piping volume of typical setups [21].

For the comparative analysis of solid ceramic breeders, a specific multiplier configuration was defined to ensure adequate neutron population, informed as a design common in a breadth of older and more recent literature [10, 21, 30]. These simulations utilized a Titanium-Beryllide multiplier (Be_{12}Ti) mixed with the ceramic breeder at a Breeder-to-Multiplier volume ratio of 1:3. To accurately approximate the macroscopic density of a ceramic pebble-bed concept, a packing factor of 62% was applied to the material definition, accounting for the inter-particle void space inherent to packed beds.

4.3.3. *Reactor Geometry and First Wall*

The First Wall (FW) material is Zirconium Alloy 4 (Zr-4: 98.2% Zr, 1.5% Sn, 0.2% Fe, 0.1% Cr). Zirconium is selected over standard ferritic-martensitic steels for the base model due to its significantly lower neutron absorption cross-section [48]. The FW thickness is set to 25 mm, an average value derived from structural integrity studies [45, 49].

The breeder blanket depth is set to an average of 500 mm, consistent with modular designs found in literature [49, 31]. The reactor triangularity is fixed at 0.55 in the base model.

4.4. *Simulation Strategy and Parametric Sweeps*

To explore the design space surrounding the Base Model, this project implements a custom, automated simulation pipeline. Leveraging the Python interfaces of both Paramak and OpenMC, an iterative script varies specific design parameters while holding the remaining Base Model attributes constant.

The workflow for each parametric sweep follows this sequence:

1. **Parameter Injection:** A target variable (e.g., Blanket Thickness, ^6Li Enrichment, or Coolant Fraction) is modified in the input dictionary.
2. **Geometry Reconstruction:** Paramak regenerates the 3D CAD model and converts it to the DAGMC h5m format.
3. **Material Assignment:** OpenMC applies the material definitions, updating isotopic fractions (e.g., adjusting the lead-lithium ratio) as required by the sweep.
4. **Transport Simulation:** The stochastic transport simulation is executed, tallying neutron interactions in the blanket volume to determine the TBR.

This methodology allows for the isolation of individual variable effects, providing clear trends on how geometric and material changes influence the overall neutronic performance of the reactor.

5. RESULTS

5.1. Material Composition and Performance

The neutronic performance of candidate liquid and solid breeder materials is summarised in Table 1 and Table 2, respectively.

Table 1: A summary of liquid breeder materials, their properties, and simulated TBR (error on each ≤ 0.005) and simulated Average Heating per Neutron (error on each ≤ 0.05).

Material	Ratio (molar %)	Li (molar %)	ρ (g/cm ³)	TBR	Heating (MeV)
Li ₁₇ Pb ₈₃	-	17.00	11.00	1.61	11.54
LiF–BeF ₂	2:1	28.57	2.04	1.15	12.92
LiF–BeF ₂ *	1:1	20.00	2.06	1.16	12.97
LiF–PbF ₂	2:3	15.38	3.55	1.23	11.63
LiF	-	50.00	2.64	1.21	13.49
Li	-	100	0.51	1.17	13.60
Li ₃ N	-	75.00	1.30	1.13	13.33
LiF–BeF ₂ –NaF	1:1:1	14.29	2.15	1.09	12.42
LiF–NaF–ZrF ₄	55:22:23	20.45	2.72	1.07	11.59
LiF–LiI ₂	83.5:16.5	50.00	3.68	1.02	10.96
LiF–LiBr–NaF	14:79:7	45.69	3.20	0.98	15.57
LiF–LiBr–NaBr	20:73:7	46.50	3.16	0.97	15.55
LiCl–PbCl ₂	1:2	12.50	4.50	0.96	9.87
LiCl–NaCl	72:28	36.00	1.52	0.91	11.04
LiF–NaF–KF	46.5:11.5:42	20.17	2.02	0.90	11.33
LiCl–BeCl ₂	1:1	20.00	1.52	0.88	10.90
LiCl–KCl	7:3	35.00	1.51	0.85	10.72
LiCl–KCl*	52:48	26.00	1.51	0.80	10.37

Among the liquid breeders, the eutectic Lithium-Lead (Li₁₇Pb₈₃) alloy demonstrates the highest TBR of 1.61. The distinctive factors to this superior performance are attributed to higher density and high lead multiplier fraction. The density arises as it is an eutectic alloy that can utilise higher packing efficiency [39, 27]. While high TBR typically correlates with increased nuclear heating due to the exothermic ⁶Li reaction, Li₁₇Pb₈₃ is a notable outlier. It exhibits lower-than-expected heating relative to its TBR because the neutron multiplication reactions in lead are endothermic.

Molten salts generally result in lower TBRs compared to liquid metals but are investigated due to their reduced Magnetohydrodynamic (MHD) effects. Of the halide salts, Fluorine-based compounds (e.g., FLiBe) perform best due to their low neutron absorption cross-sections. While Chlorine and Bromine salts were analyzed, Bromine salts exhibit high parasitic absorption, leading to significantly higher heating values (approx. 15.5 MeV) despite lower tritium production. However, it is noted that with realistic First Wall thicknesses, Bromine outperforms Chlorine due to its moderating properties. Pure lithium, LiF and LiN₃ (and to a lesser extent LiF–LiI₂) have significant Li-7 TBR contributions due to the lack of a dedicated breeder element within the compound.

Table 2: A summary of solid breeder materials, their properties, and simulated TBR (error on each ≤ 0.005) and simulated Average Heating per Neutron (error on each ≤ 0.05).

Material	Li (molar %)	ρ (g/cm ³)	TBR	Heating (MeV)
Li ₄ SiO ₄	44.44	2.40	1.47	14.88
Li ₂ TiO ₃	33.33	3.43	1.46	14.63
Li ₈ PbO ₆	53.33	4.28	1.51	15.01
Li ₈ CeO ₆	53.33	3.25	1.51	15.29
Li ₈ ZrO ₆	53.33	2.98	1.50	14.99
Li ₈ SnO ₆	53.33	3.41	1.49	15.18
Li ₆ Zr ₂ O ₇	40.00	3.56	1.49	14.79
Li ₈ GeO ₆	53.33	2.64	1.49	14.97
Li ₅ FeO ₄	50.00	2.64	1.48	14.91
Li ₅ AlO ₄	50.00	2.25	1.48	15.00
Li ₈ SiO ₆	53.33	2.20	1.48	15.03
Li ₆ ZnO ₄	54.55	2.86	1.48	14.95
Li ₄ TiO ₄	44.44	2.57	1.48	14.89
Li ₆ MnO ₄	54.55	2.50	1.48	14.91
Li ₄ GeO ₄	44.44	3.16	1.48	14.79
Li ₂ O	66.66	2.10	1.47	14.92
Li ₂ SiO ₃	33.3	2.53	1.45	14.75
Li ₂ MnO ₂	40.00	3.90	1.43	14.42
Li ₈ CoO ₆	53.33	2.47	1.43	14.69
Li ₆ CoO ₄	54.55	2.77	1.41	14.58

Table 2 indicates that solid ceramics display remarkably consistent performance compared to liquids, displayed by the low standard deviation on the average TBR: $\text{TBR} = 1.48 \pm 0.03$. This uniformity arises both from the relatively similar atomic compositions of the molecules (mostly lithium and oxygen) and the standardised multiplier configuration (Be_{12}Ti at a 1:3 ratio) defined in Section 4.3.2.

Among the specific compounds, Cerium-based ceramics achieved the highest TBR due to the low parasitic capture cross-sections of Cerium. Conversely, Cobalt-based compounds act as outliers, yielding lower TBRs despite the high multiplier fraction; this is attributed to the higher absorption and scattering cross-sections of Cobalt, which degrade the neutron economy.

5.2. Isotopic and Compositional Optimisation

Figure 5 illustrates the sensitivity of TBR to ^6Li enrichment. The dip in TBR at high percentages is attributed to the complete depletion of ^7Li , eliminating the contribution of $^7\text{Li}(n, n'\alpha)t$ reactions which are valuable for preserving the neutron population in the high-energy spectrum.

It is noted that the enrichment level of 60% used in the base model was selected *a priori* based on literature standards. However, the simulation data indicates that this value lies well within the saturation plateau, suggesting that the neutronic maximum is reached at significantly lower enrichment percentages.

The simplified study of the lead multiplier fraction in LiPb breeders is shown in Figure 6, where an approximation was taken where the density is calculated as a simple weighted average of Li and Pb. Though there was a peak at 50% lead composition, this TBR was still far lower than the TBR from $\text{Li}_{17}\text{Pb}_{83}$. These results show that density is a stronger factor affecting liquid TBR improvements than simple multiplier - breeder ratios.

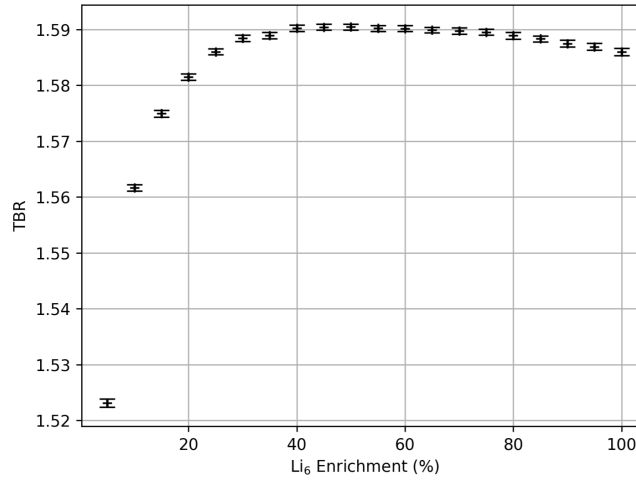


Figure 5: Tritium Breeding Ratio (TBR) as a function of ^6Li enrichment percentage. The curve is characterised by a rapid initial increase which saturates early; the “knee” of the curve appears at approximately 20%, with the TBR effectively capping at 40%. Beyond this saturation point, the returns are diminishing. Notably, a slight suppression in breeding performance is observed at near-total enrichment levels ($> 90\%$)

5.3. Impact of Coolant Selection

The influence of coolant fraction and type on breeding efficiency is detailed in Figure 7. For Helium and Nitrogen, TBR decreases monotonically as the coolant fraction increases. This behaviour indicates that these gases effectively act as voids within the breeder blanket, displacing active breeding material.

Conversely, Water (H_2O) exhibits a non-monotonic trend where TBR initially increases before declining. Notably the TBR was the very similar at 40% water coolant as when no coolant was simulated. This is attributed to the moderating properties of water, which thermalises the neutron spectrum. As shown in Figure 7b, the water coolant shifts the incident neutron population toward lower energies. Since the tritium production cross-section of ^6Li is significantly higher at thermal energies, this moderation compensates for the displacement of breeder volume at lower coolant fractions. This moderation will also reduce the neutron leakage, and so improve heating along with TBR. This then introduces the importance of considering neutron mean free path / flux attenuation and leakage within the blanket and so leads us towards the geometric studies.

5.4. Geometric Parameter Sweeps

Results from the geometric optimisation studies indicate that reactor triangularity has a negligible impact on global neutronics (Figure 8), which is expected given the uniform ring source approximation.

However, the First Wall (FW) thickness acts as a significant driver of performance. Figure 9 shows a linear decrease in TBR as FW thickness increases, caused by parasitic absorption in the structural material. A Material comparison reveals that while EUROFER [50] is the standard structural choice, Zirconium-4 alloy provides superior neutronic performance, as they have the lowest parasitic neutron absorption.

Finally, Figure 10 determines the optimal radial depth of the breeder blanket. TBR increases rapidly with thickness before saturating in the 400–600 mm range. This saturation suggests that the neutron flux is fully attenuated beyond this depth, a conclusion supported by the reaction density heatmap (Figure 10b). Increasing the breeder depth beyond 600 mm yields negligible gains in tritium production.

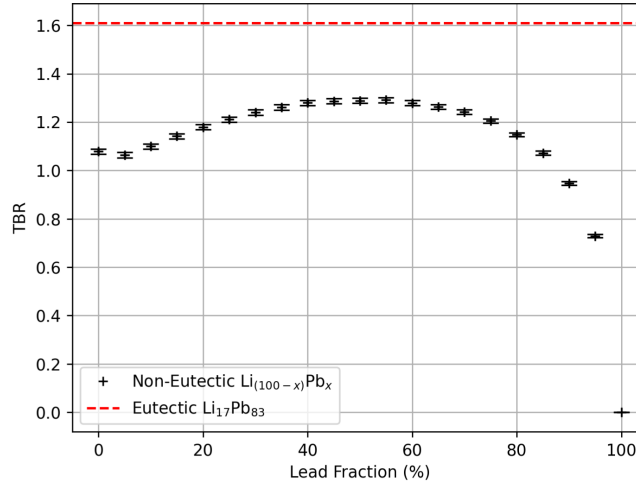


Figure 6: TBR dependence on Lead fraction in a simplified non-eutectic / constant packing factor $\text{Li}_{(100-x)}\text{Pb}_x$ breeder, where the density is calculated as a simple weighted average of Li and Pb. Breeding performance peaks at approximately 50% lead concentration, with a rapid decline observed beyond 75%. The eutectic point is marked for reference, far exceeding any non-eutectic TBR.

6. DISCUSSION

The results presented in Section 5 quantify the neutronic performance of various breeder blanket concepts. While the simulation data identifies specific maxima for Tritium Breeding Ratio (TBR), the selection of a viable blanket design for a fusion power plant requires a holistic evaluation that balances neutronic efficiency against thermal-hydraulic, structural, and economic constraints [11, 6].

6.1. TBR Optimisation Strategies

6.1.1. Isotopic and Compositional Efficiency

The study confirms that the eutectic Lithium-Lead alloy ($\text{Li}_{17}\text{Pb}_{83}$) offers superior breeding performance compared to molten salts, primarily driven by the neutron multiplication capabilities of lead [12]. However, the simulation highlights a critical physical distinction: the superior performance of the eutectic alloy over non-eutectic mixtures (Figure 6) cannot be explained by stoichiometry alone. The eutectic formation allows for a packing efficiency that results in a density approximately 15% higher than the weighted average of its constituents [44]. This increased macroscopic cross-section is the dominant factor in maximising the neutron economy.

A significant finding for reactor optimisation is the behaviour of ^6Li enrichment (Figure 5). While high enrichment is often assumed to be beneficial, our results indicate a saturation point at approximately 40% enrichment. Beyond this point, the blanket becomes “neutron-limited” rather than “target-limited.” Furthermore, enrichment levels approaching 100% actually degrade performance. This is attributed to the total depletion of ^7Li , which removes the contribution of the $^7\text{Li}(n, n'\alpha)t$ reaction [5]. Although this reaction is less frequent, it preserves the neutron population by emitting a secondary neutron, whereas the ^6Li reaction is purely absorptive. Economically, this suggests that costly high-enrichment processes are unnecessary; a moderate enrichment of 20–40% represents the optimal trade-off between TBR and fuel cycle costs [39, 27].

6.1.2. Spatial and Geometric Optimisation

The saturation of TBR at a radial depth of roughly 500 mm (Figure 10) can be validated physically by considering the neutron mean free path 8. Dominated by lead, the macroscopic

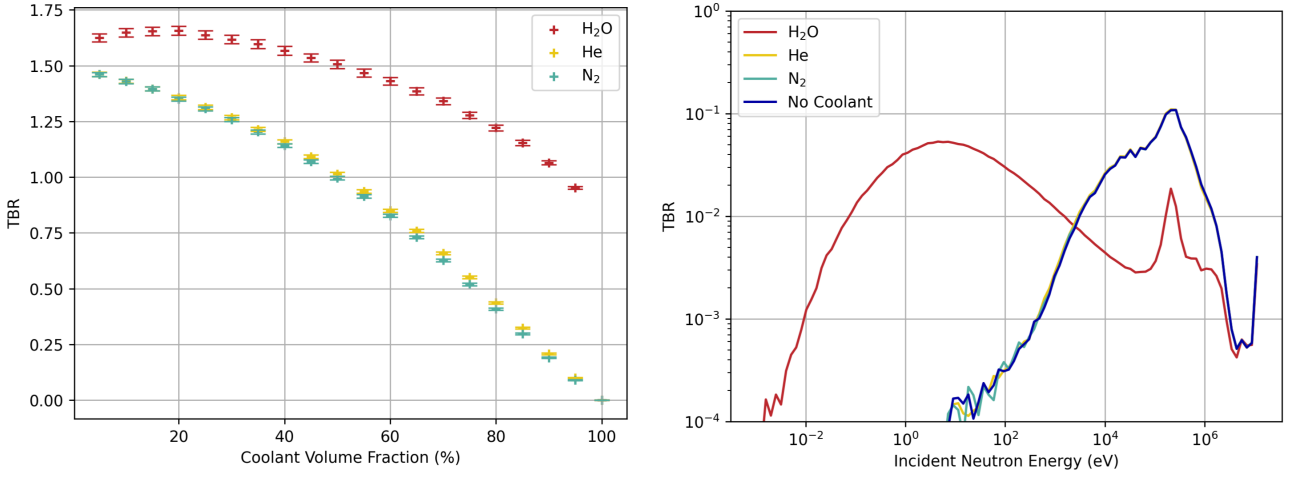


Figure 7: Impact of coolant choice on breeding performance. (a) TBR vs. Fraction of Breeder Blanket Layer replaced with Coolant (by volume). Gaseous coolants (He, N) act as voids and cause steady reductions in TBR as fractions increase. H₂O, however, has a relatively flat TBR curve up to a fraction of 0.5, even increasing TBR at low fractions. The TBR had shallower decreases at high water fractions. (b) Incident Neutron Energy spectrum for Coolant Volume Fraction = 0.33. The H₂O spectrum shows significant moderation compared to the gaseous coolants, increasing the interaction cross-section.

total cross-section (Σ_t) of the breeder mixture yields a mean free path of $\lambda \approx 5.5$ cm for 14 MeV neutrons. A 500 mm blanket therefore represents approximately 9 mean free paths, sufficient to attenuate over 99.9% of the incident flux [13]. Increasing the blanket thickness beyond this point provides no neutronic benefit and would act only as dead weight, complicating remote maintenance and increasing capital costs [20].

6.2. Consequences to Other Critical Criteria

While this study maximised TBR, the “optimal” neutronic configurations often conflict with other critical reactor criteria.

6.2.1. Structural Integrity vs. Neutronics

The use of Zirconium-4 (Zr-4) as the First Wall (FW) material yielded the highest TBR due to its low parasitic absorption cross-section (Figure 9) [48]. However, Zr-4 is generally unsuitable for the extreme neutron fluence and thermal stress of a DEMO-class reactor FW. Standard Reduced Activation Ferritic-Martensitic (RAFM) steels, such as EUROFER 97, are the engineering standard despite their higher neutron penalty [50]. Our results suggest that transitioning from an idealised Zr-4 wall to a realistic steel wall results in a TBR penalty of approximately 0.1 to 0.15. This parasitic loss must be compensated for by increasing the breeder volume or enrichment, highlighting the tight margins in fusion neutronics [6].

6.2.2. Safety and Coolant Selection

The choice of coolant represents a stark trade-off between performance and safety. Water (H₂O) was shown to enhance TBR (Figure 7) significantly better than gaseous coolants due to its moderating power, which softens the neutron spectrum into the thermal region where the ⁶Li cross-section is largest [46]. However, the use of water in a lithium-based blanket introduces the risk of violent exothermic chemical reactions in the event of a leak [32, 47]. While Helium avoids this chemical reactivity, our results show it acts effectively as a void, increasing neutron leakage

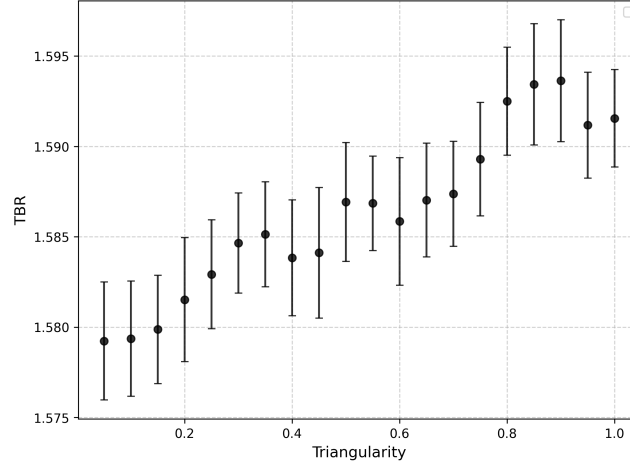


Figure 8: Sensitivity of TBR to reactor triangularity (δ). The gradient of 0.015 ± 0.001 indicates that triangularity has a negligible impact on global neutronics, consistent with the use of a uniform ring source model.

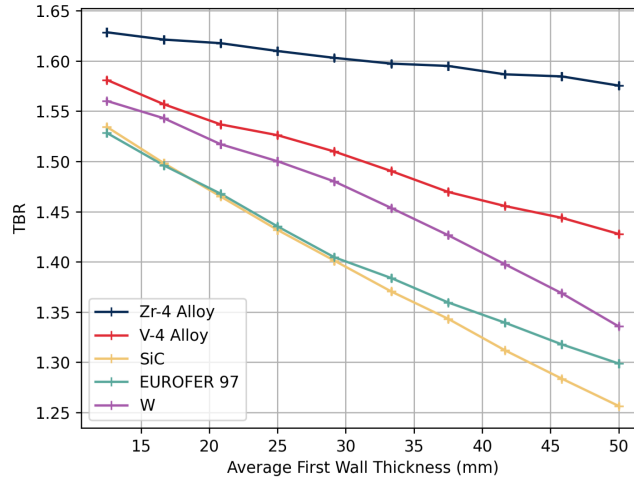


Figure 9: TBR degradation as a function of First Wall (FW) thickness for various candidate materials, including EUROFER97 [50] and Zr-4 as defined in Section 4.3.2. Zirconium-based alloys demonstrate the lowest parasitic absorption, maintaining the highest TBR profile. Error bars are included but are too small to be visible at this scale.

and degrading TBR [21]. A realistic design must therefore weigh the neutronic advantage of water moderation against the engineering complexity of double-walled piping and pressure suppression systems required to mitigate chemical hazards [31].

6.2.3. Energy Extraction (Heating)

A counter-intuitive trade-off exists regarding nuclear heating. While $\text{Li}_{17}\text{Pb}_{83}$ provided the highest TBR, it exhibited lower average heating per neutron compared to ceramic breeders (Table 1 vs Table 2). This is intrinsic to the physics of the multiplier: the $(n, 2n)$ reaction in lead is endothermic, consuming incident energy to liberate neutrons [51]. Conversely, the parasitic absorption in materials like Bromine salts yields high heating but low breeding [28]. For a commercial power plant, where electricity generation is as vital as fuel self-sufficiency, the lower energy multiplication factor of lead-lithium systems is a distinct disadvantage that must be factored into the overall plant efficiency calculations [19].

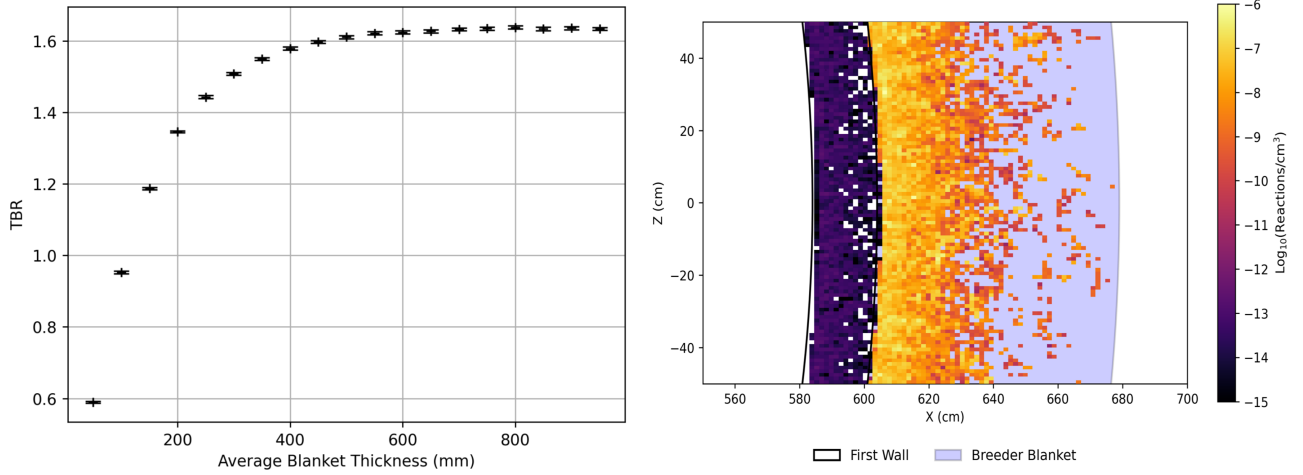


Figure 10: Analysis of Breeder Blanket radial thickness. (a) TBR saturates at approximately 500 mm, suggesting full neutron attenuation. (b) Cross-sectional heatmap of reaction density. The reaction rate drops to negligible levels beyond the $640 - 580 = 60\text{cm}$ depth into the blanket, confirming the saturation thickness observed in (a).

6.3. Model Limitations

The primary limitation of this study lies in the homogenisation of the blanket materials. By treating the breeder and coolant as a uniform mixture, the simulation neglects “streaming effects,” where neutrons escape through the void spaces of coolant channels without interaction [38]. This likely results in a slight overestimation of the TBR in our results compared to a heterogeneous model.

Furthermore, the negligible impact of triangularity on TBR (Figure 8) is likely an artifact of the simplified neutron source definition. The model utilised a uniform ring source which does not geometrically conform to the plasma shape variations associated with high triangularity [41]. In a physical tokamak, increasing triangularity compresses the plasma volume and alters the view factor of the neutrons incident on the first wall, an effect not captured by the current source definition [52].

6.4. Next Steps

Future work will focus on increasing the fidelity of the simulation model to address the limitations identified above.

- **Heterogeneous Blanket Geometry:** Utilising the advanced features of Paramak to model discrete cooling channels and pebble beds to quantify the effects of neutron streaming [41]. Additionally, factoring in packing factors for $\text{Li}_x\text{Pb}_{100-x}$ alloys could provide a more informed sweep of neutron multiplier ratios.
- **Plasma Source Definition:** Replacing the ring source with a parametric plasma source that accurately maps the spatial distribution of fusion events, allowing for a true investigation into the effects of plasma geometry [53].
- **Structural Integration:** Moving beyond the simplified structure of the few most relevant neutronics layers, and instead including a more complex structure including key components like the external toroidal field coils. This will provide a more conservative and realistic estimate of the global TBR [20, 49].

Beyond increasing model fidelity, future studies should adopt a more holistic design approach that optimises other critical criteria alongside TBR, such as nuclear heating. Coupled parameter sweep studies offer a robust framework for this multi-objective optimisation. A particular focus should be placed on minimising neutron leakage fractions; ensuring leakage approaches zero is essential for maximising the neutron economy available for breeding and heating, while simultaneously mitigating safety risks associated with external neutron radiation. Tracking this parameter is therefore recommended to ensure global design optimality.

7. CONCLUSION

This project utilised a parametric Paramak-OpenMC workflow to optimise tokamak breeder blankets for tritium self-sufficiency. By quantifying the impact of material composition and geometry, this study identified the critical design constraints required to achieve a Tritium Breeding Ratio (TBR) greater than unity.

The eutectic alloy $\text{Li}_{17}\text{Pb}_{83}$ emerged as the optimal breeder, achieving a maximum TBR of 1.61. This superior performance results from the high packing density of the eutectic phase and effective $(n, 2n)$ multiplication in lead, significantly outperforming molten salt and ceramic concepts. Crucially, this study overturns the assumption that maximal enrichment is required; results indicate TBR saturation at 40% ^6Li . Enrichment beyond this threshold degrades performance by eliminating the neutron-preserving $^7\text{Li}(n, n'\alpha)t$ reaction, suggesting that commercial reactors can operate efficiently with moderate enrichment.

Geometric parameter sweeps determined a saturation blanket thickness of 500–600 mm, governed by the neutron mean free path. Additionally, water coolant was found to enhance breeding via spectral moderation, whereas gaseous coolants increased leakage, presenting a clear trade-off between neutronic efficiency and chemical safety.

Ultimately, while these results establish robust trends, the reliance on homogenised materials and simplified sources likely overestimates TBR. Future work must therefore transition to high-fidelity heterogeneous models—incorporating discrete cooling channels and realistic plasma sources—to provide the conservative predictions necessary for the engineering design of next-generation fusion devices.

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A. SUPPLEMENTARY NEUTRONICS THEORY

A.1. Elastic Scattering Kinematics

In an elastic collision with a nucleus of mass number A , the average logarithmic energy decrement per collision (ξ) is determined by the target mass [1]:

$$\xi = 1 + \frac{(A-1)^2}{2A} \ln \left(\frac{A-1}{A+1} \right) \quad (10)$$

For Hydrogen ($A \approx 1$), $\xi = 1$, indicating that a neutron can lose all its energy in a single head-on collision. For Lead ($A \approx 208$), $\xi \approx 0.0096$, necessitating hundreds of collisions to achieve the same thermalisation.